

Personalized Content Discovery on the Netflix Prize Dataset

A four-model recommendation system with explainable Top-K recommendations

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Problem Overview

- Content discovery drives engagement on streaming platforms.
- **Data:** Netflix Prize — 100M+ ratings, 480K users, 17.7K movies, 1–5★ with timestamps.
- **Goal:** learn preferences → predict unseen ratings → generate **useful ranked Top-K** recommendations — not just minimise error.
- **Core challenge:** a ~99% sparse user–item matrix.

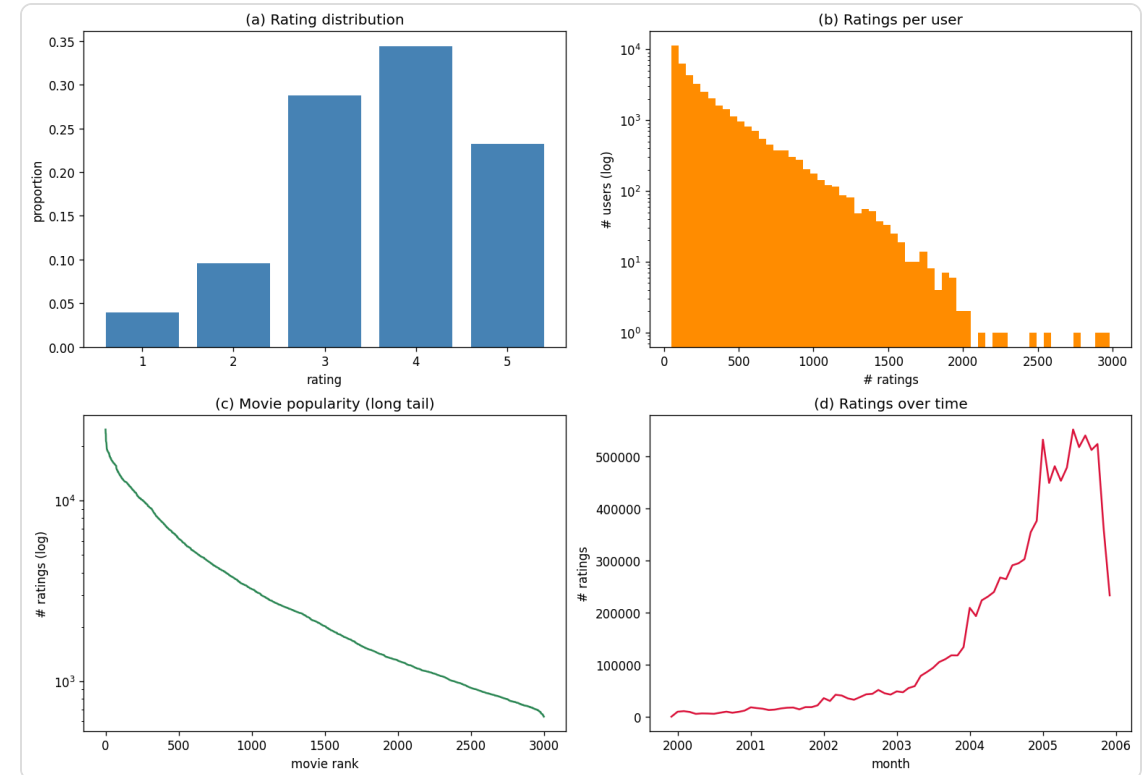
We evaluate on **both** rating accuracy (RMSE) and ranking quality (MAP@10), and study the trade-off between them.

Data & Key Insights (EDA)

- Ratings **positively skewed** (mean ≈ 3.6 ; 58% are 4–5★).
- Ratings/user: mean 267 (50–2,981); ratings/movie: mean 3,566.
- **Popularity bias**: top 20% of movies = **56%** of all ratings.
- Heavy long tail in both users and items.

Implication

Dense signal favours latent-factor models; the long tail demands attention to **coverage** for genuine discovery.



Approach

Dense, reproducible subset

Top 3,000 movies × 40,000 active users → **10.7M ratings @ 8.92% density**. Temporal 80/20 split (predict each user's most recent ratings). Relevance ≥ 3.5 .

Four models compared

- **Bias baseline** — μ + user/item biases (RMSE floor)
- **Item-based CF** — adjusted-cosine similarity (powers explanations)
- **Matrix Factorization (SVD)** — 64 latent factors, SGD on GPU
- **Neural CF** — MLP over embeddings (*innovation*)

Evaluation Methodology

- **RMSE** — rating-prediction accuracy on held-out test ratings.
- **MAP@10** — for each user: score all movies → **mask seen movies** → take Top-10 → Average Precision vs relevant (≥ 3.5) test movies.
- Also: Precision@10, Recall@10, NDCG@10, and catalog **Coverage**.
- **Temporal split** avoids the leakage of a random split — realistic, deployment-like evaluation.

8.6M

train ratings

2.1M

test ratings

40K

users evaluated

≥ 3.5

relevance threshold

Experimental Results

Model	RMSE ↓	MAP@10 ↑	NDCG@10 ↑	Coverage ↑	train (s)
Bias baseline	0.9039	0.0364	0.0731	0.015	0.7
Item-based CF	0.8838	0.0658	0.1223	0.810	1.6
Matrix Factorization (SVD)	0.8611	0.0720	0.1266	0.139	5.9
Neural CF	0.8419	0.0423	0.0817	0.351	9.7

- **RMSE ≠ ranking:** NCF wins RMSE but SVD wins MAP@10 → **SVD is the best recommender.**
- **Accuracy vs discovery:** Item-CF covers 81% of the catalog vs SVD's 14%.
- All models beat the baseline and historical Cinematch (0.9514).

Recommendation Examples & Explainability

✓ Success — User 64120 (likes Saving Private Ryan, Titanic, A Few Good Men)

Recommends **LOTR: Return of the King** — "*because you liked The Two Towers (sim 0.68) & Fellowship (sim 0.66)*" — plus Band of Brothers, Schindler's List. Hits in test set.

✗ Failure — User 2157060 (eclectic: Full Metal Jacket, Metropolis, JFK, Halloween)

Diffuse cross-genre taste → sensible-but-wrong picks (Fight Club, Citizen Kane), 0 Top-10 hits. Hard-to-rank profiles + popularity crowding.

Hit Rate@10 = 44.5% of users get ≥ 1 correct Top-10 recommendation; every pick carries a human-readable reason.

Key Insights & Future Work

Insights

- On dense data, **latent factors (SVD)** give the best accuracy–ranking blend.
- Always report **ranking metrics**, not just RMSE — they disagree.
- Popularity bias is measurable (coverage) and worth optimising for discovery.

Future work

- **Cold start:** popularity fallback, onboarding ratings, content-based **hybrid**.
- **Ranking-aware training** (BPR/WARP); **SVD++** & temporal dynamics.
- Coverage/diversity re-ranking to counter popularity bias.